

Positive Square Energy of Every Cyclomatic-Rank-Thirteen Cactus

AI-generated manuscript; no human author is claimed

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Abstract

We prove that every connected cactus G of cyclomatic rank thirteen on n vertices satisfies $s^+(G) > n$. The sharp cactus doubly nonnegative formula leaves exactly the cycle multisets $T^{12}Q$ and $T^{11}PP$, where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. Existing all-rank hostile and nonhostile one-cycle theorems close the first family for every parity of Q , and the existing all-rank two-pentagon theorem closes the second. We give the new exact frontier calculation and cite, rather than duplicate, the three rank-uniform structural proofs.

1 Sharp DNN bound and exact frontier

All graphs are finite, simple, and undirected. A cactus is connected and has only bridge or cycle blocks; its cyclomatic rank is the number of cycle blocks. For adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For an edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0, \, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [2, 1] states that if a cactus has b bridge blocks and cycle blocks $C_{\ell_1}, \dots, C_{\ell_r}$, then

$$s^-(G) \leq \kappa(G) = b + \sum_{i=1}^r \kappa(C_{\ell_i}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even,} \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases} \quad (1)$$

Set

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even,} \\ \ell \tan^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases}$$

For rank thirteen, $|E(G)| = n + 12$ and block counting gives $b + \sum_i \ell_i = n + 12$. Since $s^+(G) + s^-(G) = 2|E(G)|$, equation (1) yields

$$\boxed{\sigma(G) \geq 12 - \sum_{i=1}^{13} \epsilon_{\ell_i}.} \quad (2)$$

Proposition 1.1 (Exact rank-thirteen frontier). *The right side of (2) is nonpositive exactly for*

$$\boxed{T^{12}Q} \quad (q \geq 3, \text{ including } Q = T), \quad \boxed{T^{11}PP}.$$

Every other rank-thirteen cycle multiset has positive surplus.

Proof. For real $x \geq 3$, $\epsilon_x = x \tan^2(\pi/(2x))$ decreases strictly. Indeed, with $u = \pi/(2x)$, the derivative of $\tan^2 u/u$ has the sign of $2u - \sin u \cos u > 0$. Thus every nontriangle contributes at most

$$a := \epsilon_5 = 5 - 2\sqrt{5}.$$

The exact comparisons needed below are

$$0 < a < 1, \quad 3a < 2, \quad 2a > 1, \quad a + \epsilon_7 < 1. \quad (3)$$

After moving radicals to positive sides, the squaring certificates are $20 < 25$, $4 < 5$, $169 < 180$, and $80 < 81$. For the last inequality, $\cos(\pi/7) > 2159/2401 > 7/8$ gives $\epsilon_7 < 7/15$, while $7/15 < 2\sqrt{5} - 4 = 1 - a$ follows from $4489 < 4500$.

Let t be the number of triangles. If $t \leq 10$, monotonicity and (3) give $\sum_i \epsilon_{\ell_i} \leq 10 + 3a < 12$. If $t = 11$, the two nontriangles contribute less than one when one is even, or when their odd lengths are not both five, since then their sum is at most $a + \epsilon_7 < 1$. The unique remaining pair is PP , and $11 + 2a > 12$. If $t \geq 12$, the multiset is $T^{12}Q$ and its excess sum is $12 + \epsilon_q \geq 12$. These cases are exhaustive. The exact note and fail-closed arithmetic verifier independently freeze this classification [3]. $\square$

2 Closure of the frontier

Proposition 2.1. *Every connected cactus whose cycle blocks are twelve triangles and one cycle $Q = C_q$ satisfies $s^+(G) > |V(G)|$.*

Proof. If $q \equiv 1 \pmod{4}$, then $q \geq 5$ and the all-rank triangle–hostile-cycle theorem applies [4]. If q is even or $q \equiv 3 \pmod{4}$, the all-rank nonhostile one-cycle theorem applies [5]. These alternatives exhaust every integer $q \geq 3$, and the nonhostile theorem explicitly includes $q = 3$. Both theorems allow arbitrary shared cuts, bridge connectors, and finite tree attachments. $\square$

Proposition 2.2. *Every connected cactus whose cycle blocks are eleven triangles and two pentagons satisfies $s^+(G) > |V(G)|$.*

Proof. This is the all-rank two-pentagon theorem [6], specialized to $r = 11$. Its statement permits arbitrary cactus incidence, bridge lengths and branching, and arbitrary finite tree attachments. $\square$

Theorem 2.3. *Every connected cyclomatic-rank-thirteen cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. Outside the two families in Proposition 1.1, strict positivity follows immediately from (2). Proposition 2.1 closes $T^{12}Q$ for every parity of Q , and Proposition 2.2 closes $T^{11}PP$. The frontier classification is exhaustive. $\square$

3 Reproduction, scope, and disclosure

From the repository root, reproduce the exact frontier audit with Python 3.10 or newer:

```
python3 research/rank-thirteen-cactus-dnn-residual-frontier-verifier.py
python3 -O research/rank-thirteen-cactus-dnn-residual-frontier-verifier.py
```

Both runs report the certificate SHA-256 digest

afed9ecb78b7def1cf0daf14655730e64f52fe59a1e8a38f1b9f115b5aecce76.

The SHA-256 digest of the verifier file itself is

7da0684ff251c57a6832c5ce5077c76105af3d116eb30532f54007c2a5ec7fa9.

The verifier also rejects four hostile mutations. It certifies only the sharp-DNN frontier; positivity on the residual families comes from the cited all-rank manuscripts.

Scoped nonclaim. This paper proves only the stated result for finite simple connected cacti of cyclomatic rank thirteen. It does not claim the positive-square-energy conjecture for all graphs, a theorem for arbitrary cactus rank, or a result for non-cactus block intersections. The two-pentagon theorem makes no claim for three pentagons.

AI disclosure

This manuscript was generated and assembled by an AI coding and mathematical reasoning system from the cited repository manuscripts, note, and exact verifier. No human authorship, review, or verification is claimed. The result should be assessed from the displayed reduction and cited proof objects.

References

- [1] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.
- [2] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; `sharp-cactus-dnn/paper.tex`.
- [3] *Exact rank-thirteen cactus DNN residual frontier*, 30 July 2026; `research/rank-thirteen-cactus-dnn-residual-frontier-2026-07-30.md` and `research/rank-thirteen-cactus-dnn-residual-frontier-verifier.py`.
- [4] Companion manuscript, *Cacti with Arbitrarily Many Triangles and One Hostile Cycle*, 2026; `all-rank-triangle-hostile-cacti/paper.tex`.
- [5] *All-rank nonhostile one-cycle cactus theorem*, 30 July 2026; `research/all-rank-nonhostile-one-cycle-theorem-2026-07-30.md`.
- [6] Companion manuscript, *Positive Square Energy for Cacti with Triangles and Two Pentagons*, 2026; `all-rank-triangle-two-pentagon-cacti/paper.tex`.